

## Exercise JA

Q1. Let  $z$  be a complex number such that the imaginary part of  $z$  is nonzero and  $a = z^2 + z + 1$  is real. then  $a$  cannot take the value :

- (A)  $-1$     (B)  $\frac{1}{3}$     (C)  $\frac{1}{2}$     (D)  $\frac{3}{4}$

Ans. D

Sol. -

Q2. Let  $w = \frac{\sqrt{3} + i}{2}$  and  $P = \{w^n : n = 1, 2, 3, \dots\}$ . Further  $H_1 = \left\{z \in C : \operatorname{Re} z > \frac{1}{2}\right\}$  and  $H_2 = \left\{z \in C : \operatorname{Re} z < \frac{-1}{2}\right\}$ , where  $C$  is the set of all complex numbers. If  $z_1 \in P \cap H_1$ ,  $z_2 \in P \cap H_2$  and  $O$  represents the origin, then  $\angle z_1 O z_2 =$

- (A)  $\frac{\pi}{2}$     (B)  $\frac{\pi}{6}$     (C)  $\frac{2\pi}{3}$     (D)  $\frac{5\pi}{6}$

Ans. C, D

Sol. -

Q3. Let  $\omega$  be a complex cube root of unity with  $\omega \neq 1$  and  $P = [p_{ij}]$  be a  $n \times n$  matrix with  $p_{ij} = \omega^{i+j}$ . Then  $P^2 \neq 0$ , when  $n =$

- (A) 57    (B) 55    (C) 58    (D) 56

Ans. D, C, B

Sol. -

### Paragraph

Let  $S = S_1 \cap S_2 \cap S_3$ , where  $S_1 = \{z \in C : |z| < 4\}$ ,  $S_2 = \left\{z \in C : \operatorname{Im} \left[ \frac{z - 1 + \sqrt{3}i}{1 - \sqrt{3}i} \right] > 0\right\}$  and  $S_3 = \{z \in C : \operatorname{Re} z > 0\}$

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Q4. Area of  $S =$

- (A)  $10\pi/3$     (B)  $20\pi/3$     (C)  $16\pi/3$     (D)  $32\pi/3$

Ans. B

Sol. -

Q5.  $\min_{z \in S} |1 - 3i - z| =$

- (A)  $\frac{2 - \sqrt{3}}{2}$     (B)  $\frac{2 + \sqrt{3}}{2}$     (C)  $\frac{3 - \sqrt{3}}{2}$     (D)  $\frac{3 + \sqrt{3}}{2}$

Ans. C

Sol. -

Q6. Let complex numbers  $\alpha$  and  $\frac{1}{\alpha}$  lie on circles  $(x - x_0)^2 + (y - y_0)^2 = r^2$  and  $(x - x_0)^2 + (y - y_0)^2 = 4r^2$ , respectively. If  $z_0 = x_0 + iy_0$  satisfies the equation  $2|z_0|^2 = r^2 + 2$ , then  $|\alpha| =$

- (A)  $\frac{1}{\sqrt{2}}$  (B)  $\frac{1}{2}$  (C)  $\frac{1}{\sqrt{7}}$  (D)  $\frac{1}{3}$

Ans. C

Sol. -

Q7. Let  $z_k = \cos\left(\frac{2k\pi}{10}\right) + i \sin\left(\frac{2k\pi}{10}\right)$ ;  $k = 1, 2, \dots, 9$ .

Column-I

Column-II

P. For each  $z_k$  there exists a  $z_j$  such that  $z_k \cdot z_j = 1$

1. True

Q. There exists a  $k \in \{1, 2, \dots, 9\}$  such that  $z_1 \cdot z_k =$

2. False

$z_k$  has no solution  $z$  in the set of complex numbers.

R.  $\frac{|1 - z_1||1 - z_2| \dots |1 - z_9|}{10}$  equals

3. 1

S.  $1 - \sum_{k=1}^9 \cos\left(\frac{2k\pi}{10}\right)$  equals

4. 2

- (A) P Q R S (B) 2 1 4 3 (C) 1 2 3 4 (D) 2 1 4 3  
2 1 3 4

Ans. C

Sol. -

Q8. Column I

Column II

(A) In  $\mathbb{R}^2$ , if the magnitude of the projection vector of the vector  $\alpha\hat{i} + \beta\hat{j}$  on

(P) 1

$\sqrt{3}\hat{i} + \hat{j}$  is and  $\sqrt{3}$  if  $\alpha = 2 + \sqrt{3}\beta$  then possible value(s) of  $|\alpha|$  is (are)

(B) Let  $a$  and  $b$  be real numbers such that the function  $f(x) = \begin{cases} -3ax^2 - 2, & x < 1 \\ bx + a^2, & x \geq 1 \end{cases}$

(Q) 2

is differentiable for all  $x \in \mathbb{R}$ . Then possible value(s) of  $a$  is (are)

(C) Let  $\omega \neq 1$  be a complex cube root of unity. If

(R) 3

$(3 - 3\omega + 2\omega^2)^{4n+3} + (2 + 3\omega - 3\omega^2)^{4n+3} + (-3 + 2\omega + 3\omega^2)^{4n+3} = 0$ ,

then possible value(s) of  $n$  is (are)

(D) Let the harmonic mean of two positive real numbers  $a$  and  $b$  be 4. If  $q$  is a

(S) 4

positive real number such that a, 5, q, b is an arithmetic progression, then the value(s) of  $|q - a|$  is (are)

(T) 5

(A) (A) PQ (B) PQ (C) PQST (D) QT (B) (A) PQ (B) PS (C) PQT (D) QT

(C) (A) PQ (B) PT (C) PQS (D) QT (D) (A) TQ (B) SQ (C) PQST (D) PT

Ans. 5ef19416239ecc768b15fe47

Sol. -

Q9. For any integer k, let  $\alpha_k = \cos\left(\frac{k\pi}{7}\right) + i \sin\left(\frac{k\pi}{7}\right)$ , where  $i = \sqrt{-1}$ . The value of the

expression  $\frac{\sum_{k=1}^{12} |\alpha_{k+1} - \alpha_k|}{\sum_{k=1}^3 |\alpha_{4k-1} - \alpha_{4k-2}|}$  is

Ans. 4

Sol. -

Q10. Let  $z = \frac{-1 + \sqrt{3}i}{2}$ , where  $i = \sqrt{-1}$  and  $r, s \in \{1, 2, 3\}$ . Let  $P = \begin{bmatrix} (-z)^r & Z^{2s} \\ Z^{2s} & Z^r \end{bmatrix}$  and I be the identity matrix of order 2. Then the total number of ordered pairs (r, s) for which  $P^2 = -I$  is.

Ans. 1

Sol.  $z = \frac{-1 + \sqrt{3}i}{2} = \omega$

$$P^2 = \begin{bmatrix} (-\omega)^2 & (\omega)^{2s} \\ (\omega)^{2s} & \omega^r \end{bmatrix} \begin{bmatrix} (-\omega)^2 & (\omega)^{2s} \\ (\omega)^{2s} & \omega^r \end{bmatrix}$$

$$P^2 = -I$$

$$= \begin{bmatrix} (-\omega)^{2r} + \omega^{4s} & \omega^{2s}(\omega^2 + (-\omega)^0) \\ \omega^{2s}((-\omega)^r + \omega^r) & \omega^{2r} + \omega^{4s} \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\Rightarrow \omega^{2r} + \omega^{4s} = -1 \text{ and } \omega^{2s}(\omega^r + (-\omega)^r) = 0$$

$$\Rightarrow 1 + \omega^s + \omega^{2r} = 0 \text{ and } r \neq \text{odd}$$

$$\Rightarrow r = s = 1$$

Q11. Let  $a, b \in \mathbb{R}$  and  $a^2 + b^2 \neq 0$ . Suppose  $S = \left\{ z \in \mathbb{C} : z = \frac{1}{a + ibt}, t \in \mathbb{R}, t \neq 0 \right\}$ , where  $i = \sqrt{-1}$ . If  $z = x + iy$  and  $z \in S$ , then (x, y) lies on

(A) the circle with radius  $\frac{1}{2a}$  and centre  $\left(\frac{1}{2a}, 0\right)$  for  $a > 0, b \neq 0$

(B) the circle with radius  $-\frac{1}{2a}$  and centre  $\left(-\frac{1}{2a}, 0\right)$  for  $a < 0, b \neq 0$

(C) the x-axis for  $a \neq 0, b = 0$  (D) the y-axis for  $a = 0, b \neq 0$

Ans. A, C, D

Sol.  $x + iy = \frac{a - ibt}{a^2 + b^2t^2}$

$$\Rightarrow x = \frac{a}{a^2 + b^2t^2}, y = \frac{-bt}{a^2 + b^2t^2}$$

(C) for  $b = 0 \Rightarrow x = \frac{1}{a}, y = 0$

(D) for  $a = 0 \Rightarrow x = 0, y = \frac{-1}{bt}$

(A) Eliminating  $t$  we get

$$x^2 + y^2 - \frac{x}{a} = 0$$

Q12. Let  $a, b, x$  and  $y$  be real numbers such that  $a - b = 1$  and  $y \neq 0$ . If the complex number  $z = x + iy$  satisfies  $\text{Im} \frac{az + b}{z + 1} = y$ , then which of the following is(are) possible value(s) of  $x$ ?

(A)  $1 + \sqrt{1 + y^2}$  (B)  $-1 + \sqrt{1 - y^2}$  (C)  $1 - \sqrt{1 + y^2}$  (D)  $-1 - \sqrt{1 - y^2}$

Ans. B, D

Sol.  $\text{Im} \left( \frac{a(x + iy) + b}{x + iy + 1} \right) = y$

$$\text{Im} \left( \frac{(ax + b) + iay}{(x + 1) + iy} \right) = y$$

$$\frac{ay(x + 1) - y(ax + b)}{(x + 1)^2 + y^2} = y$$

$$a - b = (x + 1)^2 + y^2$$

$$1 = (x + 1)^2 + y^2$$

$$(x + 1)^2 = 1 - y^2$$

$$(x + 1) = \pm \sqrt{1 - y^2}$$

$$x = -1 \pm \sqrt{1 - y^2}$$

Q13. For a non-zero complex number  $z$  let  $\arg(z)$  denote the principal argument with  $-\pi < \arg(z) \leq \pi$ . Then, which of the following statement(s) is (are) FALSE ?

(A)  $\arg(-1 - i) = \frac{\pi}{4}$ , where  $i = \sqrt{-1}$

(B) The function  $f: \mathbb{R} \rightarrow (-\pi, \pi]$  defined by  $f(t) = \arg(-1 + it)$  for all  $t \in \mathbb{R}$ , is continuous at all points of  $\mathbb{R}$ , where  $i = \sqrt{-1}$

(C) For any two non-zero complex numbers  $z_1$  and  $z_2$ ,  $\arg\left(\frac{z_1}{z_2}\right) - \arg(z_1) + \arg(z_2)$  is an integer multiple of  $2\pi$

(D) For any three given distinct complex numbers  $z_1, z_2$  and  $z_3$ , the locus of the point  $z$  satisfying the condition  $\arg \left( \frac{(z - z_1)(z_2 - z_3)}{(z - z_3)(z_2 - z_1)} \right) = \pi$ , lies on a straight line

Ans. A, B, D

Sol. (A)  $\text{Arg}(-1 - i) = -\frac{3\pi}{4}$

(B)  $f(t) = \text{Arg}(-1 + it)$

$$\begin{cases} \pi - \tan^{-1} t & t \geq 0 \\ -(\pi + \tan^{-1} t) & t < 0 \end{cases}$$

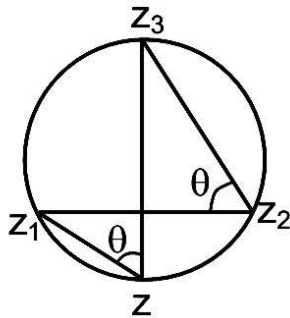
Its is discontinuous at  $t = 0$

(C)  $\text{Arg}\left(\frac{z_1}{z_2}\right) = \text{Arg } z_1 - \text{Arg } z_2$

$$\text{Arg}\left(\frac{z_1}{z_2}\right) = \text{Arg } z_1 - \text{Arg } z_2 + 2n\pi$$

so the expression becomes  $2n\pi$

(D)



$$\text{Arg}\left(\frac{(z - z_1)(z_2 - z_3)}{(z - z_3)(z_2 - z_1)}\right) = \pi$$

If is circle

Q14. Let  $s, t, r$  be non-zero complex numbers and  $L$  be the set of solutions  $z = x + iy$  ( $x, y \in \mathbb{R}, i = \sqrt{-1}$ ) of the equation  $sz + t\bar{z} + r = 0$ , where  $\bar{z} = x - iy$ . Then, which of the following statement(s) is (are) TRUE ?

(A) If  $L$  has exactly one element, then  $|s| \neq |t|$

(B) If  $|s| \neq |t|$ , then  $L$  has infinitely many elements

(C) The number of elements in  $L \cap \{z : |z - 1 + i| = 5\}$  is at most 2

(D) If  $L$  has more than one element, then  $L$  has infinitely many elements

Ans. D, C, A

Sol.  $az + t\bar{z} + r = 0, \bar{z} = x - iy, s\bar{z} + \bar{t}z + r = 0(1) + (2)(t + \bar{s})\bar{z} + (s + \bar{t})z + (r + \bar{r}) = 0(t + \bar{s})\bar{z} + (s + \bar{t})z + (r + \bar{r}) = 0$

$$\frac{t + \bar{s}}{t - \bar{s}} = \frac{\bar{t} + s}{s - \bar{t}} = \frac{r + \bar{r}}{r - \bar{r}} |t| = |s| \quad \bar{t}r - \bar{t}\bar{r} + sr - s\bar{r} = sr + s\bar{r} - \bar{t}r - \bar{t}\bar{r} \quad 2\bar{t}r = :$$

$$\bar{t}r = s\bar{r}$$

$$|\bar{t}||r| = |s||r|$$

$$|t| = |s|$$

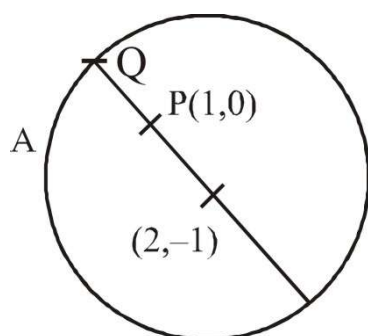
If  $|t| = |s|$ , lines will be parallel for sure but it may not be coincident

For option (C) if element of set L represent line, then this line and given circle can hav

- Q15. Let S be the set of all complex numbers  $z$  satisfying  $|z - 2 + i| \geq \sqrt{5}$  if the complex number  $z_0$  is such that  $\frac{1}{|z_0 - 1|}$  is the maximum of the set  $\left\{ \frac{1}{|z - 1|} : z \in S \right\}$  then the principal argument of  $\frac{4 - z_0 - \bar{z}_0}{z_0 - \bar{z}_0 + 2i}$  is
- (A)  $\frac{\pi}{4}$  (B)  $-\frac{\pi}{2}$  (C)  $\frac{\pi}{2}$  (D)  $\frac{3\pi}{4}$

Ans. B

Sol.



Set S will be set of all points being outside (on) circle A whose center is  $(2, -1)$  & radi

$|z - 1|$  = distance of  $z$  from  $(1, 0)$  Q is the required point when  $\frac{1}{|z - 1|}$  is maximum.  $Q(x_0, y_0) \Rightarrow z_0$

- Q16. Let  $\omega \neq 1$  be a cube root of unity. Then the minimum of the set  $\{|a + b\omega + c\omega^2|^2 : a, b, c \text{ distinct non-zero integers}\}$  equals -

Ans. -

Sol.

$$\begin{aligned}
 |a + b\omega + c\omega^2|^2 &= (a + b\omega + c\omega^2) \overline{(a + b\omega + c\omega^2)} \\
 &= (a + b\omega + c\omega^2) (a + b\omega^2 + c\omega) \\
 &= a^2 + b^2 + c^2 - ab - bc - ca \\
 &= \frac{1}{2} [(a - b)^2 + (b - c)^2 + (c - a)^2] \\
 &\geq \frac{1 + 1 + 4}{2} = 3 \text{ (when } a = 1, b = 2, c = 3)
 \end{aligned}$$

- Q17. Let S be the set of all complex numbers  $z$  satisfying  $|z^2 + z + 1| = 1$ . Then which of the following statements is/are TRUE ?

(A)  $\left| z + \frac{1}{2} \right| \leq \frac{1}{2}$  for all  $z \in S$  (B)  $|z| \leq 2$  for all  $z \in S$  (C)  $\left| z + \frac{1}{2} \right| \geq \frac{1}{2}$  for all  $z \in S$

(D) The set S has exactly four elements

Ans. B, C, D

Sol.  $|z^2 + z + 1| = 1$

$$\Rightarrow \left| \left( z + \frac{1}{2} \right)^2 + \frac{3}{4} \right| = 1$$

$$\Rightarrow \left| \left( z + \frac{1}{2} \right)^2 + \frac{3}{4} \right| \leq \left| z + \frac{1}{2} \right|^2 + \frac{3}{4}$$

$$\Rightarrow 1 \leq \left| z + \frac{1}{2} \right|^2 + \frac{3}{4} \Rightarrow \left| z + \frac{1}{2} \right|^2 \geq \frac{1}{4}$$

$$\Rightarrow \left| z + \frac{1}{2} \right| \geq \frac{1}{2}$$

Also  $|(z^2 + z) + 1| = 1 \Rightarrow |z^2 + z| - 1 \leq 1$

$$\Rightarrow |z^2 + z| - 1 \leq 1$$

$$\Rightarrow |z^2 + z| \leq 2$$

$$\Rightarrow ||z^2| - |z|| \leq |z^2 + z| \leq 2$$

$$\Rightarrow |r^2 - r| \leq 2$$

$$\Rightarrow r = |z| \leq 2; \forall z \in S$$

Also we can always find root of the equation

$$z^2 + z + 1 = e^{i\theta}; \forall \theta \in \mathbb{R}$$

Hence set 'S' is infinite

Q18.

Ans. -

Sol. -

Q19. For any complex number  $w = c + id$ , let  $\arg(w) \in (-\pi, \pi]$  where  $i = \sqrt{-1}$ . Let  $\alpha$  and  $\beta$  be real numbers such that for all complex numbers  $z = x + iy$  satisfying

$$\arg\left(\frac{z + \alpha}{z + \beta}\right) = \frac{\pi}{4}, \text{ the ordered pair } (x, y) \text{ lies on the circle } x^2 + y^2 + 5x - 3y + 4 = 0$$

Then which of the following statements is (are) TRUE ?

(A)  $\alpha = -1$     (B)  $\alpha\beta = 4$     (C)  $\alpha\beta = -4$     (D)  $\beta = 4$

Ans. D, B

Sol.  $\arg\left(\frac{z + \alpha}{z + \beta}\right) = \frac{\pi}{4}; \alpha, \beta \text{ are real numbers}$

$$\Rightarrow \arg(z + \alpha) - \arg(z + \beta) = \frac{\pi}{4}$$

$$\Rightarrow \arg(x + \alpha + iy) - \arg(x + \beta + iy) = \frac{\pi}{4}$$

$$\Rightarrow \tan^{-1}\left(\frac{y}{\alpha + x}\right) - \tan^{-1}\left(\frac{y}{\beta + x}\right) = \frac{\pi}{4}$$

$$\Rightarrow \tan^{-1}\left[\frac{\frac{y}{\alpha + x} - \frac{y}{\beta + x}}{1 + \left(\frac{y}{\alpha + x}\right)\left(\frac{y}{\beta + x}\right)}\right] = \frac{\pi}{4}$$

$$\Rightarrow \frac{\beta y + xy - \alpha y - xy}{\alpha\beta + x(\alpha + \beta) + x^2 + y^2} = \tan \frac{\pi}{4} = 1$$

$$\beta y - \alpha y = \alpha\beta + x(\alpha + \beta) + x^2 + y^2$$

$$\Rightarrow x^2 + y^2 + x(\alpha + \beta) + y(\alpha - \beta) + \alpha\beta = 0$$

$$x^2 + y^2 + 5x - 3y + 4 = 0$$

$$\Rightarrow \alpha + \beta = 5 \quad \alpha\beta = 4$$

$$\alpha - \beta = -3$$

$$\alpha = 1$$

$$\beta = 4$$

(B), (D) are correct

Q20. Let  $\theta_1, \theta_2, \dots, \theta_{10}$  be positive valued angles (in radian) such that  $\theta_1 + \theta_2 + \dots + \theta_{10} = 2\pi$ . Define the complex numbers  $z_1 = e^{i\theta_1}$ ,  $z_k = z_{k-1}e^{i\theta_k}$  for  $k = 2, 3, \dots, 10$ , where  $i = \sqrt{-1}$ . Consider the statements P and Q given below:

$$P: |z_2 - z_1| + |z_3 - z_2| + \dots + |z_{10} - z_9| + |z_1 - z_{10}| \leq 2\pi$$

$$Q: |z_2^2 - z_1^2| + |z_3^2 - z_2^2| + \dots + |z_{10}^2 - z_9^2| + |z_1^2 - z_{10}^2| \leq 4\pi$$

Then,

(A) P is TRUE and Q is FALSE      (B) Q is TRUE and P is FALSE

(C) both P and Q are TRUE      (D) both P and Q are FALSE

Ans. C

Sol.  $\theta_1 + \theta_2 + \dots + \theta_{10} = 2\pi$

$$z_1 = e^{i\theta_1}, z_2 = z_1 e^{i\theta_2}, z_3 = z_2 e^{i\theta_3} \dots \dots \dots$$

$$= |z_{k-1}| |e^{i\theta_k} - 1| = \left| 2 \sin \left( \frac{\theta_k}{2} \right) \right|$$

$$|z_k^2 - z_{k-1}^2| = |z_{k-1}^2 e^{i2\theta_k} - z_{k-1}^2| = |z_{k-1}^2| |e^{i2\theta_{k-1}} - 1| = |2 \sin \theta_k|$$

**Statement P**

$$\begin{aligned} & |z_2 - z_1| + |z_3 - z_2| + \dots + |z_{10} - z_9| + |z_1 - z_{10}| \\ & \left| 2 \sin \frac{\theta_2}{2} \right| + \left| 2 \sin \frac{\theta_3}{2} \right| + \dots + \left| 2 \sin \frac{\theta_{10}}{2} \right| + \left| 2 \sin \left( \frac{\theta_2 + \dots + \theta_{10}}{2} \right) \right| \\ & \left| 2 \sin \frac{\theta_2}{2} \right| + \left| 2 \sin \frac{\theta_3}{2} \right| + \dots + \left| 2 \sin \frac{\theta_{10}}{2} \right| + \left| 2 \sin \left( \frac{2\pi - \theta_1}{2} \right) \right| \\ & = \left| 2 \sin \frac{\theta_1}{2} \right| + \left| 2 \sin \frac{\theta_2}{2} \right| + \dots + \left| 2 \sin \frac{\theta_{10}}{2} \right| \end{aligned}$$

Maximum value will occur when  $\theta_1 = \theta_2 \dots = \theta_{10} = \frac{2\pi}{10}$

$$\text{LHS}_{\max} = 10 \left( 2 \sin \frac{\pi}{10} \right) = 20 \sin (18^\circ) = 20 \left( \frac{\sqrt{5}-1}{4} \right) \approx 6.18 < 2\pi$$

$\Rightarrow$  P is true

**Statement Q**

$$\text{LHS} = |2 \sin \theta_1| + |2 \sin \theta_2| + \dots + |2 \sin \theta_{10}|$$

Maximum Value will occur when  $\theta_1 = \theta_2 = \dots = \theta_{10} = \frac{\pi}{5}$ .

$$\text{LHS} = 10 \left( 2 \sin \frac{\pi}{5} \right) = 20 \sin 36^\circ < 4\pi$$

$\Rightarrow$  Statement Q is true.

Q21. Let  $z$  be a complex number with non-zero imaginary part. If  $\frac{2 + 3z + 4z^2}{2 - 3z + 4z^2}$  is a real number, then the value of  $10|z|^2$  is \_\_\_\_\_.

Ans. 5

Sol. Let  $w = \frac{4z^2 + 3z + 2}{4z^2 - 3z + 2} = 1 + \frac{6z}{4z^2 - 3z + 2}$

$$\Rightarrow w = 1 + \frac{6}{2 \left( 2z + \frac{1}{z} \right) - 3}$$

$$\therefore w \in \mathbb{R} \text{ then } 2z + \frac{1}{z} \in \mathbb{R}$$

$$\Rightarrow 2z + \frac{1}{z} = 2\bar{z} + \frac{1}{\bar{z}}$$

$$\Rightarrow 2(z - \bar{z}) - \frac{z - \bar{z}}{|z|^2} = 0$$

$$\Rightarrow (z - \bar{z}) \left( 2 - \frac{1}{|z|^2} \right) = 0$$

$\because z \neq \bar{z}$  given

$$\text{So } |z|^2 = \frac{1}{2}$$

Q22. Let  $\bar{z}$  denote the complex conjugate of a complex number  $z$  and let  $i = \sqrt{-1}$ . In the set of complex numbers, the number of distinct roots of the equation  $\bar{z} - z^2 = i(\bar{z} + z^2)$  is \_\_\_\_\_.

Ans. 4.00

Sol.  $\bar{z} - z^2 = i(\bar{z} + z^2)$

$$\Rightarrow (1 - i)\bar{z} = (1 + i)z^2$$

$$\Rightarrow \frac{(1 - i)}{(1 + i)}\bar{z} = z^2$$

$$\Rightarrow \left( -\frac{2i}{2} \right) \bar{z} = z^2$$

$$\therefore z^2 = -i\bar{z}$$

Let  $z = x + iy$ ,

$$\therefore (x^2 - y^2) + i(2xy) = -i(x - iy)$$

$$\text{so, } x^2 - y^2 + y = 0 \quad \text{_____ (1)}$$

$$\text{and } (2y + 1)x = 0 \quad \text{_____ (2)}$$

$$\therefore x = 0 \text{ or } y = -\frac{1}{2}$$

Case I : When  $x = 0$

$$\therefore (1) \Rightarrow y(1 - y) = 0 \therefore y = 0, 1$$

$$\therefore (0, 0), (0, 1)$$

Case II : When  $y = -\frac{1}{2}$

$$\therefore (1) \Rightarrow x^2 - \frac{1}{4} - \frac{1}{2} = 0 \Rightarrow x^2 = \frac{3}{4} \Rightarrow x = \pm \frac{\sqrt{3}}{2}$$

$$\therefore \left( \frac{\sqrt{3}}{2}, -\frac{1}{2} \right), \left( -\frac{\sqrt{3}}{2}, -\frac{1}{2} \right)$$

$$\Rightarrow \text{Number of distinct 'z' is equal to 4.}$$

Q23. Let  $\bar{z}$  denote the complex conjugate of a complex number  $z$ . If  $z$  is a non-zero complex number for which both real and imaginary parts of  $(\bar{z})^2 + \frac{1}{z^2}$  are integers, then which of the following is/are possible value(s) of  $|z|$  ?

(A)  $\left( \frac{43 + 3\sqrt{205}}{2} \right)^{\frac{1}{4}}$  (B)  $\left( \frac{7 + \sqrt{33}}{4} \right)^{\frac{1}{4}}$  (C)  $\left( \frac{9 + \sqrt{65}}{4} \right)^{\frac{1}{4}}$  (D)  $\left( \frac{7 + \sqrt{13}}{6} \right)^{\frac{1}{4}}$

Ans. A

Sol. Let  $z = r(\cos \theta + i \sin \theta)$

$$\bar{z}^2 + \frac{1}{z^2} = r^2(\cos 2\theta - i \sin 2\theta)$$

$$+ \frac{1}{r^2}(\cos 2\theta - i \sin 2\theta)$$

$$= \left( r^2 + \frac{1}{r^2} \right) (\cos 2\theta - i \sin 2\theta)$$

How

$$\left. \begin{aligned} \left( r^2 + \frac{1}{r^2} \right) \cos 2\theta &= \lambda \Rightarrow \left( r^4 + \frac{1}{r^4} + 2 \right) \cos^2 2\theta = \lambda^2 \\ \left( r^2 + \frac{1}{r^2} \right) \sin 2\theta &= \mu \Rightarrow \left( r^4 + \frac{1}{r^4} + 2 \right) \sin^2 2\theta = \mu^2 \end{aligned} \right\} \Rightarrow r^4 + \frac{1}{r^4} + 2 = \lambda^2 + \mu^2$$

So,  $r^4 + \frac{1}{r^4} \in I$

(A)  $r^4 = \frac{43 + 3\sqrt{205}}{2} \Rightarrow \frac{1}{r^4} = \frac{2(43 - 3\sqrt{205})}{1849 - 1845} = \frac{43 - 3\sqrt{205}}{2} \Rightarrow r^4 + \frac{1}{r^4} = 43 \in I$

(B)  $r^4 = \frac{7 + \sqrt{33}}{4} \Rightarrow \frac{1}{r^4} = \frac{4(7 - \sqrt{33})}{49 - 33} = \frac{7 - \sqrt{33}}{4} \Rightarrow r^4 + \frac{1}{r^4} \notin I$

(C)  $r^4 = \frac{9 + \sqrt{65}}{4} \Rightarrow \frac{1}{r^4} = \frac{4(9 - \sqrt{65})}{81 - 65} = \frac{9 - \sqrt{65}}{4} \Rightarrow r^4 + \frac{1}{r^4} \notin I$

(D)  $r^4 = \frac{7 + \sqrt{13}}{6} \Rightarrow \frac{1}{r^4} = \frac{6(7 - \sqrt{13})}{49 - 13} = \frac{6(7 - \sqrt{13})}{36} = \frac{7 - \sqrt{13}}{6} \Rightarrow r^4 + \frac{1}{r^4} \notin I$

Q24. Let  $A = \left\{ \frac{1967 + 1686i \sin \theta}{7 - 3i \cos \theta} : \theta \in R \right\}$ . If A contains exactly one positive integer  $n$ , then the value of  $n$  is \_\_\_\_\_.

Ans. 281

Sol.  $A = \frac{1967 + 1686i \sin \theta}{7 - 3i \cos \theta}$

$$= \frac{281(7 + 6i \sin \theta)}{7 - 3i \cos \theta} \times \frac{7 + 3i \cos \theta}{7 + 3i \cos \theta}$$

$$= \frac{281(49 - 18 \sin \theta \cos \theta + i(21 \cos \theta + 42 \sin \theta))}{49 + 9 \cos^2 \theta}$$

for positive integer

$\text{Im}(A) = 0$

$$21 \cos \theta + 42 \sin \theta = 0$$

$$\tan \theta = \frac{-1}{2}; \sin 2\theta = \frac{-4}{5}, \cos^2 \theta = \frac{4}{5}$$

$$\begin{aligned} \operatorname{Re}(A) &= \frac{281(49 - 9 \sin 2\theta)}{49 + 9 \cos^2 \theta} \\ &= \frac{281(49 - 9 \times \frac{-4}{5})}{49 + 9 \times \frac{4}{5}} = 281 (+ve \text{ integer}) \end{aligned}$$

Q25. Let  $z$  be a complex number satisfying  $|z|^3 + 2z^2 + 4\bar{z} - 8 = 0$ , where  $\bar{z}$  denotes the complex conjugate of  $z$ . Let the imaginary part of  $z$  be non-zero.

Match each entry in List-I to the correct entries in List-II.

| List-I                                    | List-II |
|-------------------------------------------|---------|
| (P) $ z ^2$ is equal to                   | 12      |
| (Q) $ z - \bar{z} ^2$ is equal to         | 4       |
| (R) $ z ^2 +  z + \bar{z} ^2$ is equal to | 8       |
| (S) $ z + 1 ^2$ is equal to               | 10      |
|                                           | 7       |

$$(A) (P) \rightarrow (1); (Q) \rightarrow (3); (R) \rightarrow (5); (S) \rightarrow (4)$$

$$(B) (P) \rightarrow (2); (Q) \rightarrow (1); (R) \rightarrow (3); (S) \rightarrow (5)$$

$$(C) (P) \rightarrow (2); (Q) \rightarrow (4); (R) \rightarrow (5); (S) \rightarrow (1)$$

$$(D) (P) \rightarrow (2); (Q) \rightarrow (3); (R) \rightarrow (5); (S) \rightarrow (4)$$

Ans. 650ae509adadd6012e37fe0e

$$\text{Sol. } |z|^3 + 2z^2 + 4\bar{z} - 8 = 0 \quad \text{_____} (1)$$

taking conjugate

$$|z|^3 + 2\bar{z}^2 + 4z - 8 = 0 \quad \text{_____} (2)$$

$$(1) - (2)$$

$$2(z^2 - \bar{z}^2) + 4(\bar{z} - z) = 0$$

$$(z - \bar{z})(z + \bar{z}) + 2(\bar{z} - z) = 0$$

$$(z - \bar{z})(z + \bar{z} - 2) = 0$$

$\therefore z - \bar{z} \neq 0$  (Otherwise Imag part of  $z$  will be zero)

$$\therefore z + \bar{z} - 2 = 0$$

$$x + iy + x - iy - 2 = 0$$

$$2x - 2 = 0$$

$$x = 1$$

∴ z will be of the form  $z = 1 + iy$

$$|z| = \sqrt{1 + y^2}; z^2 = 1 - y^2 + 2iy; \bar{z} = 1 - iy$$

Substitute in equation (1)

$$(1 + y^2)^{\frac{3}{2}} + 2(1 - y^2 + 2iy) + 4(1 - iy) - 8 = 0$$

$$(1 + y^2)^{\frac{3}{2}} + 2 - 2y^2 + 4 - 8 + i(4y - 4y) = 0$$

$$(1 + y^2)^{\frac{3}{2}} + (-2y^2 - 2) = 0$$

$$(1 + y^2)^{\frac{3}{2}} - 2(1 + y^2) = 0$$

$$(1 + y^2) \left\{ \sqrt{1 + y^2} - 2 \right\} = 0$$

$$1 + y^2 = 4$$

$$y^2 = 3$$

$$(P) |z|^2 = 1 + y^2$$

$$= 4$$

$$(Q)$$

$$= |1 + iy - 1 + iy|^2$$

$$= |2iy|^2$$

$$= 4y^2$$

$$= 4 \cdot 3$$

$$= 12$$

$$(R)$$

$$= 4 + |1 + iy + 1 - iy|^2$$

$$= 4 + 4$$

$$= 8$$

$$(S) |z + 1|^2 = |1 + iy + 1|^2$$

$$= |2 + iy|^2$$

$$= 4 + 3$$

$$= 7$$

Q26. Let  $A_1, A_2, A_3, \dots, A_8$  be the vertices of a regular octagon that lie on a circle of radius 2. Let P be a point on the circle and let  $PA_i$  denote the distance between the points P and  $A_i$  for  $i = 1, 2, \dots, 8$ . If P varies over the circle, then the maximum value of the product  $PA_1 \cdot PA_2 \cdot \dots \cdot PA_8$ , is

Ans. 512

Sol.  $z_r = 2e^{i\left(\frac{2\pi r}{8}\right)} \quad r = \{0, 1, 2, \dots, 7\}$

$$PA_1 \cdot PA_2 \cdot \dots \cdot PA_8$$

$$= |z - z_1| |z - z_2| \dots |z - z_8|$$

$$= |(z - z_1)(z - z_2) \dots (z - z_8)| \quad \text{_____} (1)$$

Also

$$z^8 - 2^8 = (z - z_1)(z - z_2) \dots (z - z_8) \quad \{|z| = 2\}$$

$$\Rightarrow |z^8 - 2^8| = |(z - z_1)(z - z_2) \dots (z - z_8)|$$

by triangle inequality

$$|z^8 + (-2^8)| \leq |z|^8 + 2^8$$

$$\leq 2^8 + 2^8$$

$$\leq 2^9$$

$$\text{So } \max (PA_1 \cdot PA_2 \dots PA_8) = 2^9$$

$$= 29$$

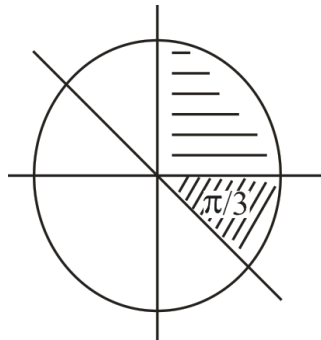
$$\text{Ans.} = 512$$

Q27. Let  $S_1 = \{z \in C : |z| \leq 5\}$ ,  $S_2 = \left\{z \in C : \operatorname{Im} \left( \frac{z+1-\sqrt{3}i}{1-\sqrt{3}i} \right) \geq 0\right\}$  and  $S_3 = \{z \in C : \operatorname{Re}(z) \geq 0\}$ . Then the area of the region  $S_1 \cap S_2 \cap S_3$  is :

(A)  $\frac{125\pi}{24}$  (B)  $\frac{125\pi}{4}$  (C)  $\frac{125\pi}{12}$  (D)  $\frac{125\pi}{6}$

Ans. C

Sol.



$$S_1 : x^2 + y^2 \leq 5^2, S_2 : \sqrt{3}x + y \geq 0, S_3 : x \geq 0$$

$$\frac{z+1-i\sqrt{3}}{1-\sqrt{3}i} = \frac{((x+1) + i(y-\sqrt{3}))(1+i\sqrt{3})}{1+3}$$

$$\operatorname{Im}(z) = (x+1)\sqrt{3} + (y-\sqrt{3}) = \sqrt{3}x + y$$

$$S_1 \cap S_2 \cap S_3 = \frac{1}{4}\pi 5^2 + \frac{1}{2}5^2 \frac{\pi}{3} = \frac{125\pi}{12}$$

## Exercise C1

---

Q1.  $\sin^{-1} \left\{ \frac{1}{i}(z-1) \right\}$ , where  $z$  nonreal, can be the angle of a triangle if

- (A)  $\operatorname{Re}(z) = 1, \operatorname{Im}(z) = 2$     (B)  $\operatorname{Re}(z) = 1, 0 < \operatorname{Im}(z) \leq 1$     (C)  $\operatorname{Re}(z) + \operatorname{Im}(z) = 0$   
(D) none of these

Ans. B

Sol. -

Q2. Let  $A, B, C$  represent the complex numbers  $z_1, z_2, z_3$  respectively on the complex plane. If the circumcentre of the triangle  $ABC$  lies at the origin, then the orthocentre is represented by the complex number :

- (A)  $z_1 + z_2 - z_3$     (B)  $z_2 + z_3 - z_1$     (C)  $z_3 + z_1 - z_2$     (D)  $z_1 + z_2 + z_3$

Ans. D

Sol. -

Q3. If  $(a + ib)^5 = \alpha + i\beta$  then  $(b + ia)^5$  is equal to

- (A)  $\beta + i\alpha$     (B)  $\alpha + i\beta$     (C)  $\beta + i\alpha$     (D)  $-\alpha - i\beta$

Ans. A

Sol. -

Q4. If  $|z| = \max \{|z-1|, |z+1|\}$  then

- (A)  $|z + \bar{z}| = \frac{1}{2}$     (B)  $z + \bar{z} = 1$     (C)  $|z + \bar{z}| = 1$     (D)  $z \in \phi$

Ans. D

Sol. -

Q5. If  $|z_1 - 1| < 1, |z_2 - 2| < 2, |z_3 - 3| < 3$  then  $|z_1 + z_2 + z_3|$

- (A) is less than 6    (B) is more than 3    (C) is less than 12    (D) lies between 6 and 12

Ans. C

Sol. -

Q6. The expression  $\left[ \frac{1 + i \tan \alpha}{1 - i \tan \alpha} \right] - \frac{1 + i \tan n \alpha}{1 - i \tan n \alpha}$  when simplified reduces to :

- (A) zero    (B)  $2 \sin n \alpha$     (C)  $2 \cos n \alpha$     (D) none

Ans. A

Sol. -

Q7. The equation of the radical axis of the two circles represented by the equations,  $|z - 2| = 3$  and  $|z - 2 - 3i| = 4$  on the complex plane is :

- (A)  $3iz - 3i\bar{z} - 2 = 0$  (B)  $3iz - 3i\bar{z} + 2 = 0$  (C)  $iz - i\bar{z} + 1 = 0$  (D)  $2iz - 2i\bar{z} + 3 = 0$

Ans. B

Sol. -

Q8. If  $z = (3 + 7i)(p + iq)$  where  $p, q \in \mathbb{I} - \{0\}$ , is purely imaginary then minimum value of  $|z|^2$  is

- (A) 0 (B) 58 (C)  $\frac{3364}{3}$  (D) 3364

Ans. D

Sol. -

Q9. Let  $z_r$  ( $1 \leq r \leq 4$ ) be complex numbers such that  $|z_r| = \sqrt{r+1}$  and  $|30z_1 + 20z_2 + 15z_3 + 12z_4| = k|z_1z_2z_3 + z_2z_3z_4 + z_3z_4z_1 + z_4z_1z_2|$ . Then the value of  $k$  equals

- (A)  $|z_1z_2z_3|$  (B)  $|z_2z_3z_4|$  (C)  $|z_3z_4z_1|$  (D)  $|z_4z_1z_2|$

Ans. D

Sol. -

Q10. Consider the equation  $10z^2 - 3iz - k = 0$ , where  $z$  is a complex variable and  $i^2 = -1$ . Which of the following statements is True?

- (A) For all real positive numbers  $k$ , both roots are pure imaginary.  
 (B) For negative real numbers  $k$ , both roots are pure imaginary.  
 (C) For all pure imaginary numbers  $k$ , both roots are real and irrational.  
 (D) For all complex numbers  $k$ , neither root is real.

Ans. B

Sol. -

Q11. If  $z^3 + (3 + 2i)z + (-1 + ia) = 0$  has one real root, then the value of 'a' lies in the interval ( $a \in \mathbb{R}$ )

- (A)  $(-2, -1)$  (B)  $(-1, 0)$  (C)  $(0, 1)$  (D)  $(1, 2)$

Ans. B

Sol. -

Q12. If  $|z| = 1$  and  $|\omega - 1| = 1$  where  $z, \omega \in \mathbb{C}$ , then the largest set of values of  $|2z - 1|^2 + |2\omega - 1|^2$  equals

- (A) [1, 9]    (B) [2, 6]    (C) [2, 12]    (D) [2, 18]

Ans. D

Sol. -

Q13. If  $\text{Arg}(z + a) = \frac{\pi}{6}$  and  $\text{Arg}(z - a) = \frac{2\pi}{3}$ ;  $a \in \mathbb{R}^+$ , then

- (A)  $z$  is independent of  $a$     (B)  $|a| = |z + a|$     (C)  $z = a \text{ Cis } \frac{\pi}{6}$     (D)  $z = a \text{ Cis } \frac{\pi}{3}$

Ans. D

Sol. -

Q14. Number of imaginary complex numbers satisfying the equation,  $z^2 = \bar{z}2^{1-|z|}$  is

- (A) 0    (B) 1    (C) 2    (D) 3

Ans. C

Sol. -

Q15. If  $\cos \theta + i \sin \theta$  is a root of the equation  $x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_{n-1} x + a_n = 0$  then the value of  $\sum_{r=1}^n a_r \cos r\theta$  equals (where all coefficient are real)

- (A) 0    (B) 1    (C) -1    (D) None

Ans. C

Sol. -

Q16. Given  $f(z)$  = the real part of a complex number  $z$ . For example,  $f(3 - 4i) = 3$ . If a  $\hat{I}$   $N, n \in \mathbb{N}$  then the value of  $\sum_{n=1}^{6a} \log_2 \left| f\left((1 + i\sqrt{3})^n\right) \right|$  has the value equal to

- (A)  $18a^2 + 9a$     (B)  $18a^2 + 7a$     (C)  $18a^2 - 3a$     (D)  $18a^2 - a$

Ans. D

Sol. -

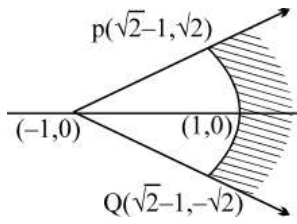
Q17. All complex numbers ' $z$ ' which satisfy the relation  $|z - |z + 1|| = |z + |z - 1||$  on the complex plane lie on the

- (A) line  $y = 0$  or an ellipse with foci  $(-1, 0)$  and  $(1, 0)$   
 (B) radical axis of the circles  $|z - 1| = 1$  and  $|z + 1| = 1$     (C) circle  $x^2 + y^2 = 1$   
 (D) line  $x = 0$  or on a line segment joining  $(-1, 0)$  to  $(1, 0)$

Ans. D

Sol. -

Q18. The locus of  $z$  which lies in shaded region is best represented by



- (A)  $z : |z + 1| > 2, |\arg(z + 1)| < \pi/4$     (B)  $z : |z - 1| > 2, |\arg(z - 1)| < \pi/4$   
 (C)  $z : |z + 1| < 2, |\arg(z + 1)| < \pi/2$     (D)  $z : |z - 1| < 2, |\arg(z - 1)| < \pi/2$

Ans. A

Sol. -

Q19. If  $w = \alpha + i\beta$  where  $\beta \neq 0$  and  $z \neq 1$ , satisfies the condition that  $\frac{w - \bar{w}z}{1 - z}$  is purely real, then the set of values of  $z$  is

- (A)  $\{z : |z| = 1\}$     (B)  $\{z : z = \bar{z}\}$     (C)  $\{z : z \neq 1\}$     (D)  $\{z : |z| = 1, z \neq 1\}$

Ans. D

Sol. -

Q20. If  $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2$  then

- (A)  $\frac{z_1}{z_2}$  is purely real    (B)  $\frac{z_1}{z_2}$  is purely imaginary    (C)  $z_1 \bar{z}_2 + z_2 \bar{z}_1 = 0$   
 (D)  $\arg \frac{z_1}{z_2}$  may be equal to  $\frac{\pi}{2}$

Ans. D

Sol. -

Q21. If  $2 \cos \theta = x + \frac{1}{x}$  and  $2 \cos \phi = y + \frac{1}{y}$ , then

- (A)  $x^n + \frac{1}{x^n} = 2 \cos(n\theta)$     (B)  $\frac{x}{y} + \frac{y}{x} = 2 \cos(\theta - \phi)$     (C)  $xy + \frac{1}{xy} = 2 \cos(\theta - \phi)$   
 (D) None of these

Ans. C, B, A

Sol. -

Q22. The loci of a point  $P(z)$  in the complex plane satisfying  $\left|z + \frac{1}{z}\right| = 2$  are two circles  $C_1$  and  $C_2$ . These circles

- (A) have centres on real axis.    (B) cut each other orthogonally.    (C) are congruent.  
 (D) have exactly two common tangents.

Ans. D, B, C

Sol. -

Q23. If  $1, \omega, \omega^2, \dots, \omega^{n-1}$  are the  $n$ th roots of unity, then  $(2 - \omega)(2 - \omega^2) \dots (2 - \omega^{n-1})$  equals :

(A)  $2^n - 1$  (B)  ${}^nC_1 + {}^nC_2 + \dots + {}^nC_n$  (C)  $[{}^{2n+1}C_0 + {}^{2n+1}C_1 + \dots + {}^{2n+1}C_n]^{1/2} - 1$

(D) None of these

Ans. A, B, C

Sol.  $1 + x + x^2 + \dots + x^{n-1} = (x - \omega)(x - \omega^2) \dots (x - \omega^{n-1})$

Put  $x = 2$

$$(2 - \omega)(2 - \omega^2) \dots (2 - \omega^{n-1}) = 1 + 2 + \dots + 2^{n-1}$$

$$= \frac{2^n - 1}{2 - 1} = 2^n - 1$$

$$= {}^nC_0 + {}^nC_1 + \dots + {}^nC_n$$

(C) We know,

$${}^{2n+1}C_0 + {}^{2n+1}C_1 + \dots + {}^{2n+1}C_n = \frac{2^{2n+1}}{2}$$

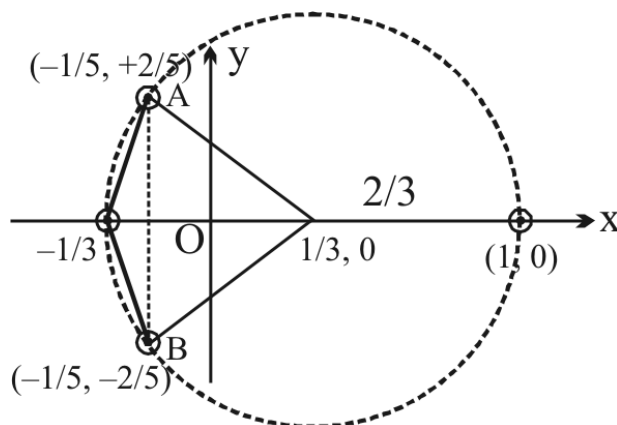
$$= 2^{2n}$$

Q24. Which of the following represents a point on an argands' plane, equidistant from the roots of the equation  $(z + 1)^4 = 16z^4$ ?

(A)  $(0, 0)$  (B)  $\left(-\frac{1}{3}, 0\right)$  (C)  $\left(\frac{1}{3}, 0\right)$  (D)  $\left(0, \frac{2}{\sqrt{5}}\right)$

Ans. C

Sol.



$$\left(\frac{z+1}{z}\right)^4 = 16 \quad \text{P} \quad \frac{z+1}{z} = 2 \text{ or } -2 \text{ or } 2i \text{ or } -2i$$

$$\text{Roots are } 1; -\frac{1}{3}; \left(-\frac{1}{5} - \frac{2}{5}i\right) \text{ and } \left(-\frac{1}{5} + \frac{2}{5}i\right)$$

Note that  $\left(-\frac{1}{3}, 0\right)$  and  $(1, 0)$  are equidistant from  $\left(\frac{1}{3}, 0\right)$

and since it lies on the perpendicular bisector of AB, it will be equidistant from A and B also.

Alternatively:  $|z + 1| = 2|z|$

$$(z + 1)(\bar{z} + 1) = 4(z\bar{z}) \dots(1)$$

This is the equation of circle with centre  $(1/3, 0)$  which is equidistant from the root of the equation.

Q25. Let  $z_1, z_2, z_3$  are the coordinates of the vertices of the triangle  $A_1A_2A_3$ . Which of the following statements are equivalent :

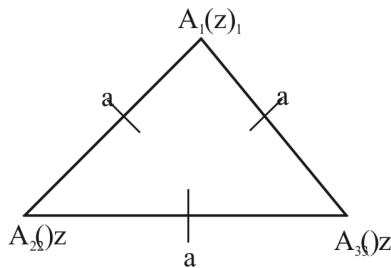
(A)  $A_1A_2A_3$  is an equilateral triangle.

(B)  $(z_1 + \omega z_2 + \omega^2 z_3)(z_1 + \omega^2 z_2 + \omega z_3) = 0$ , where  $\omega$  is the imaginary cube root of unity.

$$(C) \frac{z_2 - z_1}{z_3 - z_2} = \frac{z_3 - z_2}{z_1 - z_3} \quad (D) \begin{vmatrix} 1 & 1 & 1 \\ z_1 & z_2 & z_3 \\ z_2 & z_3 & z_1 \end{vmatrix} = 0$$

Ans. A, B, C, D

Sol.



$$\frac{z_2 - z_1}{a} = \frac{z_3 - z_1}{a} e^{\frac{i\pi}{3}}$$

$$z_2 - z_1 = (z_3 - z_1) e^{\frac{i\pi}{3}} \dots(1)$$

$$\text{Similarly, } z_3 - z_2 = (z_1 - z_3) e^{\frac{i\pi}{3}} \dots(2)$$

divide (1) by (2)

$$\frac{z_2 - z_1}{z_3 - z_2} = \frac{z_3 - z_1}{z_1 - z_3}$$

(B) (D)

For equilateral  $\Delta$ , we know

$$z_1^2 + z_2^2 + z_3^2 - z_1z_2 - z_2z_3 - z_1z_3 = 0$$

- Q26. If  $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_{n-1}$  are the imaginary  $n^{\text{th}}$  roots of unity then the product  $\prod_{r=1}^{n-1} (i - \alpha_r)$  (where  $i = \sqrt{-1}$ ) can take the value equal to :
- (A) 0    (B) 1    (C) i    (D)  $(1 + i)$

Ans. A, B, C, D

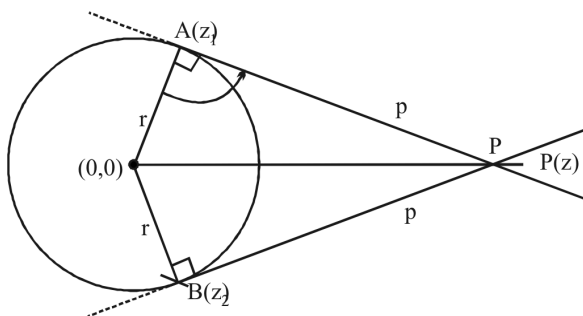
Sol.  $\frac{z^n - 1}{z - 1} = (z - \alpha_1)(z - \alpha_2)\dots\dots(z - \alpha_{n-1})$   
 put  $z = i$

$$\prod_{r=1}^{n-1} (i - \alpha_r) = \frac{i^n - 1}{i - 1} = \begin{cases} 0 & \text{if } n = 4k \\ 1 & \text{if } n = 4k + 1 \\ 1 + i & \text{if } n = 4k + 2 \\ i & \text{if } n = 4k + 3 \end{cases}$$

- Q27. Let tangents at  $A(z_1)$  and  $B(z_2)$  are drawn to the circle  $|z| = 2$ . Then which of the following is/are correct?
- (A) The equation of tangent at A is given by  $\frac{z}{z_1} + \frac{\bar{z}}{\bar{z}_1} = 2$ .
- (B) If tangents at  $A(z_1)$  and  $B(z_2)$  intersect at  $P(z_p)$ , then  $z_p = \frac{2z_1z_2}{z_1 + z_2}$ .
- (C) Slope of tangent at  $A(z_1)$  is  $\frac{1}{i} \left( \frac{z_1 + \bar{z}_1}{z_1 - \bar{z}_1} \right)$ .
- (D) If points  $A(z_1)$  and  $B(z_2)$  on the circle  $|z| = 2$  are such that  $z_1 + z_2 = 0$ , then tangents intersect at  $\frac{\pi}{2}$

Ans. A, B, C

Sol.



(A)  $\arg \left( \frac{z - z_1}{0 - z_1} \right) = \frac{\pi}{2}$

$\Rightarrow \frac{z - z_1}{z_1}$  is purely imaginary.

So,  $\frac{z_1 - z}{z_1} + \frac{\bar{z}_1 - \bar{z}}{\bar{z}_1} = 0$

$$\Rightarrow \frac{z}{z_1} + \frac{\bar{z}}{\bar{z}_1} = 2$$

(B) By applying rotation, we get (AP = p)

$$\frac{z - z_1}{0 - z_1} = \frac{p}{r} e^{i\frac{\pi}{2}} \quad \dots(1)$$

$$\text{Also } \frac{0 - z_2}{z - z_2} = \frac{r}{p} e^{i\frac{\pi}{2}} \quad \dots(2)$$

$$\therefore \text{ On multiplying (1) and (2), we get } z_p = \frac{2z_1z_2}{z_1 - z_2}$$

(C) As equation of tangent at A(x<sub>1</sub>, y<sub>1</sub>) is

$$xx_1 + yy_1 = 4$$

$$\therefore \text{ Slope of tangent } = \frac{-x_1}{y_1} = \frac{-2x_1}{2y_1} = -i \left( \frac{z_1 + \bar{z}_1}{z_1 - \bar{z}_1} \right) = \frac{1}{i} \left( \frac{z_1 + \bar{z}_1}{z_1 - \bar{z}_1} \right)$$

(D) Clearly tangents are parallel lines.

(As A(z<sub>1</sub>) and B(z<sub>2</sub>) are ends of diameter of circle.)